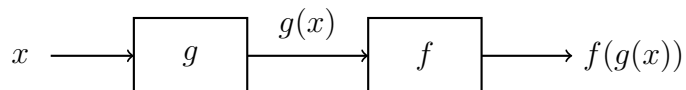


Composing Functions from Symbolic Rules

Evaluating a composition at a number, building the composed rule from symbolic definitions, deciding whether the order matters, and confirming inverses by composition

The composition machine. $(f \circ g)(x)$ is another way of writing $f(g(x))$. The output of the *inner* rule becomes the input of the *outer* rule, so the inner rule always runs first. In the picture below no numbers are shown — it is only the wiring; the rules and inputs come from each problem.



Directions.

- Replace the outer rule's input variable with the *entire* inner rule, kept in parentheses, before you simplify anything.
- Expand and combine like terms so the final rule is in simplest form.
- Show the substitution step — that step is what is being graded, not just the last line.

1. Let $f(x) = x + 4$ and $g(x) = 3x$. Send the input 2 through g first, then send that output through f . Find $(f \circ g)(2)$.

Answer: _____

2. Let $f(x) = 2x - 1$ and $g(x) = x^2$. Fill in the middle column by running each input through f , then fill in the last column by running that result through g . The last column is $(g \circ f)(x)$.

x	$f(x)$	$g(f(x))$
0		
2		
3		

Answer: _____

3. Let $f(x) = x^2 + 1$ and $g(x) = x - 3$. Write the rule for $(f \circ g)(x)$, expanded and simplified.

Answer: _____

4. Keep the same two rules, $f(x) = x^2 + 1$ and $g(x) = x - 3$, but reverse the order. Write the rule for $(g \circ f)(x)$, simplified.

Answer: _____

5. Let $f(x) = x^2 - 2x$ and $g(x) = x + 5$. Write $(f \circ g)(x)$ as a polynomial in simplest form. (The inner rule goes into *both* places where f shows its input.)

Answer: _____

6. Let $f(x) = \frac{1}{x-2}$ and $g(x) = x + 2$. Write $f(g(x))$ as a single fraction in simplest form.

Answer: _____

$$(f \circ g)(x) = \underline{\hspace{2cm}}$$

$$(g \circ f)(x) = \underline{\hspace{2cm}}$$

Answer: _____

9. Let $f(x) = 3x + 7$ and $g(x) = \frac{x-7}{3}$. Composition is the test for inverses: show that $f(g(x))$ simplifies to x *and* that $g(f(x))$ simplifies to x . Both directions are needed before you may call the two rules inverses.

Answer: _____

10. Let $h(x) = \frac{1}{x^2 + 6}$. Find rules f and g , neither of them the identity rule x , so that $h(x) = f(g(x))$. Then compose your two rules back together and explain, in a sentence, how that composition shows your choice works.